



DEPARTMENT OF ELECTRICAL ENGINEERING

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Professor

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To whom it may concern.

Kevin Doshi, who is a student pursuing a B.Tech degree in Computer Engineering, **Sardar Patel Institute of Technology, Mumbai** was a research intern who worked under my supervision at the Dept. of Electrical Engineering IIT Bombay between **19th January 2024 and 20th June 2024**. The broad area in which he worked was **“Creating a central and flexible repository for power system data”**.

The tasks carried out by Kevin Doshi are listed below:

- Conducted in-depth research on component interrelations to ensure seamless integration and proper field inclusion across all project modules.
- Designed a scalable database schema for components like transmission lines and optimized queries to enhance data retrieval performance and handle large datasets efficiently.
- Implemented MongoDB pipelines for real-time monitoring and logging of database changes, ensuring audit trails and data tracking.
- Developed a role-based access control (RBAC) system and an advanced email recovery-based authentication using NextAuth to secure access to sensitive project features.
- Crafted an intuitive user interface and home page for improved user experience and streamlined navigation, integrating third-party APIs to extend project functionality.
- Refactored and documented code for improved readability and maintainability, while authoring comprehensive project documentation and a user guide.
- Deployed the project and its documentation on Vercel to enable open-source collaboration and easy access for developers.

Kevin has gained proficiency in technologies like NextJS, tailwind CSS, MongoDB, Firebase, shadcn UI toolkit, nodemailer and next-auth package. For version control of the developed packages, he used git and GitHub. I found that Kevin was a diligent, sincere and hardworking student. I wish him well in his future endeavours.

(Prof. A. M. Kulkarni)